Model Dataflow Architecture

Repo-aligned extraction, feature engineering, and module mapping for market regime detection plus RL allocation

1 End-to-End Diagram

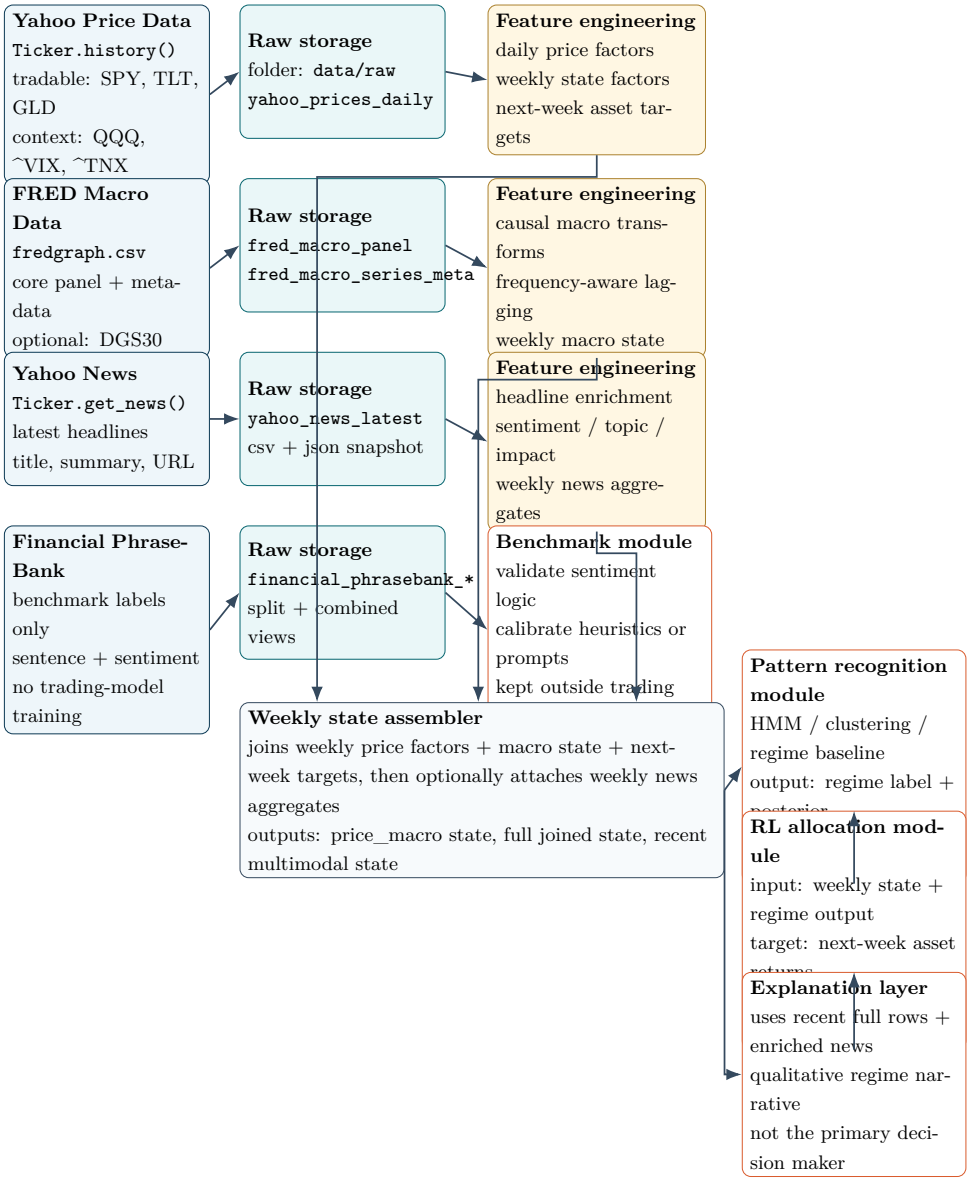

Figure 1: Data extraction and feature mapping into the regime detector, RL allocator, and explanation modules.

2 Source-to-Module Mapping

2.1 Yahoo Finance Price Data

- Extraction: `yfinance.Ticker.history()` on SPY, TLT, GLD, QQQ, ^VIX, ^TNX
- Raw schema: date, symbol, open, high, low, close, adj_close, volume
- Processed outputs: `market_features_daily.csv`, `market_features_weekly.csv`, `weekly_asset_targets.csv`

- Model use: regime core state, RL state variables, and next-period return targets.

2.2 FRED Macro Data

- Extraction: repo macro fetch pipeline with default preset `core`, using FRED-format identifiers plus series-specific fallbacks where needed.
- Core series: DFF, DGS3MO, DGS10, T10Y3M, T10YIE, NFCI, BAMLH0A0HYM2, VIXCLS, DTWEXBGS, WRESBAL, ICSA, UNRATE, CPIAUCSL, CFNAI, UMCSENT, PERMIT
- Optional duration extension: `core_plus_duration` adds `DGS30`.
- Processed output: `macro_features_weekly.csv` with transforms driven by `frequency` and `suggested_transfor`
- Model use: economic regime block for pattern recognition and macro context for RL.

2.3 Yahoo Finance News

- Extraction: `yfinance.Ticker.get_news()` on SPY, QQQ, TLT, GLD
- Raw schema: title, summary, published_at, publisher, canonical_url, related_tickers
- Processed outputs: `news_events_enriched.csv` and `news_features_weekly.csv`; current build leaves both empty because the raw Yahoo news snapshot is absent.
- Model use: recent-window multimodal regime experiments, event overlays, and explanation support.

2.4 Financial PhraseBank

- Extraction: Hugging Face archive download when the optional benchmark dataset is refreshed.
- Raw schema: sentence, label
- Model use: benchmark only for sanity-checking the sentiment pipeline; not present in the current working tree.

3 Weekly State Outputs

Output file	Contents	Main downstream use
`model_state_weekly_price_macro.csv`	Price features + macro features + weekly asset targets (625 rows, 75 columns)	Main historical regime training table and first RL table
`model_state_weekly_full.csv`	Price + macro + targets + nullable news features (625 rows, 90 columns)	Unified schema for integration code
`model_state_weekly_recent_full.csv`	Recent rows where price, macro, and news all exist (currently 0 rows, 90 columns)	Multimodal demos, case studies, and explanation outputs once the raw news snapshot is refreshed

4 Recommended FRED Core

Theme	Series	Why it belongs in the state
Policy and curve	DFF, DGS3MO, DGS10, T10Y3M	Policy stance, front-end rates, long-end rates, inversion risk
Inflation and risk pricing	T10YIE, BAMLH0A0HYM2, VIXCLS, DTWEXBGS	Inflation expectations, credit stress, volatility, dollar tightness
Liquidity and labor	WRESBAL, ICSA, UNRATE	System liquidity, weekly labor stress, slower labor confirmation
Activity and demand	CPIAUCSL, CFNAI, UMCSENT, PERMIT	Realized inflation, broad activity, consumer tone, housing lead signal

5 DGS30 Recommendation

Keep the default historical panel at the 16-series `core` preset. Add `DGS30` only when the team wants extra long-duration information for `TLT` sensitivity or term-premium interpretation. The repo now supports:

- `python3 scripts/fetch_fred_macro_panel.py -preset core`
- `python3 scripts/fetch_fred_macro_panel.py -preset core_plus_duration`

6 Causal Alignment Rules

- Decision calendar: weekly (`W-FRI`).
- Daily price and daily FRED series use the last available value in the decision week.
- Weekly FRED series are lagged by one week before joining.
- Monthly FRED series are lagged by one month before joining.
- News events are timestamped, converted to `week_end`, and aggregated within each week.
- Next-week asset returns stay shifted forward and are not part of the current feature row.

7 External References

- Yahoo Finance via `yfinance`: https://ranaroussi.github.io/yfinance/index.html
- `Ticker.history()`: https://ranaroussi.github.io/yfinance/reference/api/yfinance.Ticker.history.html
- `Ticker.get_news()`: https://ranaroussi.github.io/yfinance/reference/api/yfinance.Ticker.get_news.html
- FRED API overview: https://fred.stlouisfed.org/docs/api/fred/overview.html
- Financial PhraseBank: https://huggingface.co/datasets/takala/financial_phrasebank